Estimating the cross-linguistic prevalence of the Sonority Sequencing Principle in adult and child-directed speech

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Keywords: sonority, syllable, language acquisition, phonology, bayesian

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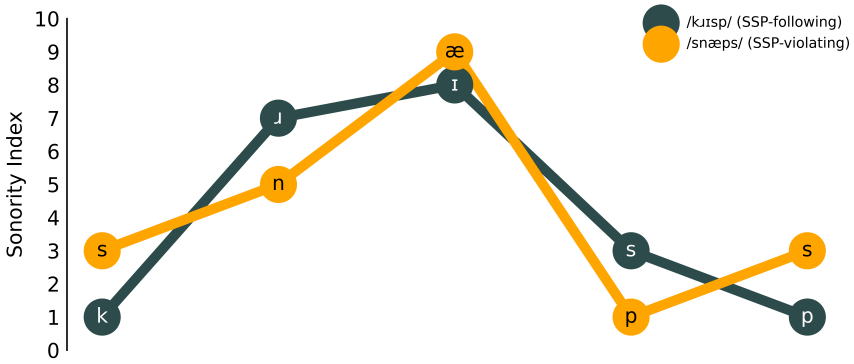
Syllables are a privileged linguistic unit in speech processing during the earliest stages of language acquisition (Bertoncini et al., 1988; Eimas, 1999; Jusczyk & Derrah, 1987). For instance, during as early as their first days of life humans are able to discriminate between word-forms based on the number of syllables, instead of relying on the number of phonemes or the duration of the produced utterance (Bijeljac-Babic et al., 1993). Newborns can also track the co-ocurrence of syllable-sized groups of speech sounds to chunk the continuous speech stream into word-like units (Fló et al., 2019, 2022), an ability that will later give them access to the precursors of a lexicon (Saffran et al., 1996; Saffran & Kirkham, 2018). Given the relatively limited experience newborns have accumulated with their native language, their ability to identify syllables in speech is remarkable. To date, the mechanism behind such an early speech processing skill are unclear.

Previous work points to newborns being sensitive to certain constraints to syllabic structure (e.g. Bertoncini & Mehler, 1981; Gomez et al., 2014; Moon & Fifer, 1990). One of them is the Sonority Sequencing Principle (SSP). The SSP constrains the order of the segments inside a syllable based a phonological property name sonority, which indexes the perceptual salience of an individual segment relative to neighbouring ones (see Albert & Nicenboim, 2022; S. Parker, 2008; S. G. Parker, 2002; Jany et al., 2007 for a discussion on the specific acoustic correlates of sonority). Sonority is conceived as an ordinal feature of individual speech sound (e.g., Clements, 1990 suggests a hierarchy such as *Obstruents* < *Nasals* < *Liquids* < *Glides* < *Vowels*), which is oftentimes represented as a numerical scale. For methodological purposes, we will adopt the numerical scale proposed by S. Parker (2008), ranging from 1 (lowest sonority) to 8 (highest sonority). The SSP states that each syllable displays a unique peak of sonority in the nucleus, and that sonority rises monotonically from the syllable margins toward the nucleus (S. G. Parker, 2002; Selkirk, 1984). Following a numeric scale of sonority, a syllable can be

evaluated as following or violating the SSP by examining its sonority contour, i.e., the sequence of sonority values assigned to each segment in the syllable (Blevins, 1995; Clements, 1990). Figure 1 illustrates the case of two English word-forms that follow and violate the SSP, respectively. The English word "crisp" /kɹɪsp/ complies with the SSP, with stop consonants /k/ and /p/ (lowest sonority) at the periphery of the syllable, /ɹ/ and /s/ (medium sonority) surrounding the syllabic nucleus, and /ɪ/ (highest sonority) at the centre of the syllable. A word like "skirt" /skɪrt/, on the other hand, violates the SSP, as its sonority contour does not increase monotonically towards the syllabic nucleus.

Figure 1

Examples of SSP-following (English word "crisp", /kɹɪsp/) and SSP-violating (English word "snaps", /snæps/) syllables according to the Sonority Index.



Evidence from experimental studies suggests that sonority—and particularly the SSP—play an important role in language processing and production in both spoken (Berent et al., 2007, 2011; Berent & Lennertz, 2010; Chen et al., 2024; Miozzo & Buchwald, 2013) and signed languages (Perlmutter, 1993; Sandler, 1993; Stone et al., 2018). The SSP has also been shown to play a role during language acquisition. Gomez et al. (2014) found newborns that hearing SSP-following syllables (e.g., "bran", "fros") elicits different brain haemodynamic responses compared to SSP-violating syllables (e.g., "lbif", "rdos"). This suggests that newborns, who have not yet accumulated extensive experience with the phonological contraints of their native language, are already sensitive to violations of the SSP. Such early bias toward certain syllabic

structures would provide humans with an available mechanism to detect syllables in the continuous speech stream at the earliest stages of language acquisition.

The SSP has been long claimed to be a linguistic universal, on the grounds of SSP-following syllables being preferred over SSP-violating syllables across most languages (e.g., Greenberg, 1978; Kreitman, 2012). For instance, S. Parker (2017) collected evidence from 275 experimental works on a typologically diverse list of 75 distinct languages, and found that 63% of the studies suggest that sonority plays a relevant role for accounting for the results, 22% provide inconclusive evidence, and 15% provide negative evidence. Experimental evidence also supports that sonority constrictions are part of some language-non-specific knowledge that humans possess: listeners prefer SSP-following over SSP-violating consonant clusters, even when both types of structure are unattested in the native language (Berent et al., 2011; Berent & Lennertz, 2010). The universality of the SSP would place this principle as one of the few cues for speech segmentation present from birth to virtually *all* humans, before language-specific constraints to sequences of speech sounds (e.g., phonotactic rules) are available.

The status of the SSP as a linguistic universal is nonetheless still a matter of debate, as violations of this principle are common across many languages. For instance, Yin et al. (2023) reported at least one violation of the SSP in around half of the 496 languages they examined. Onset consonant clusters of the type initial `fricative-stop consonant` (e.g., /st/, as in "Stein" in German) or its inverse counterpart in coda `stop consonant-fricative` (e.g., /ks/, as in "clicks" in English) are attested in many languages (Kawasaki-Fukumori et al., 1997). Some languages are known for allowing fairly more complex consonant clusters at syllabic onset whic violate the SSP. This is the case of Russian, which allows for clusters up to three consonants at syllabic onset (e.g., "взгляд" /vzglyad/ [glance]) (Halle & Jones, 1971; Itô, 1982) or Georgian, in which onset consonant clusters like /md/ are attested (e.g., "მდინარე" /ʻmdi.nä.ɾi/) (Crouch et al., 2023). Recent work on previously neglected languages

also suggests to the proportion of SSP-violating consonant clusters may be higher than prior work had anticipated in some languages (e.g., Sun & Billington, 2024 in Lelepa; Al Tamimi & Al Shboul, 2013 in Modern Standard Arabic). In light of these findings, the SSP cannot be taken as a hard linguisitic universal, but rather as an overall preference toward certain linguistic structures that that virtually all languages display to a varying extent (Frisch, 2015; Prince & Smolensky, 2004). Under this account, the possibility of the SSP being universally available as a cue for syllabification at birth becomes a quantitative question: how frequently do syllables in different languages abide by the SSP?

To address this question, we used Wiktionary database, an open-source, free database of lexical entries (Contributors to Wikimedia projects, 2026). As of June 10th 2026, Wiktionary contains 10,050,751 lexical entries from 8,244 languages contributed by 4,463,976 total users. We collected all 7,859 available phonological transcriptions from all languages in the Glottolog database (Nordhoff & Hammarström, 2011) that were present in Wiktionary, resulting in a sample of 139 languages from 17 distinct language families. For each language in the dataset, we derived a syllabic repertoire by syllabifying the available phonological transcriptions, when possible. In a first analysis, we estimated how many of the available syllables in the syllabic repertoire of each language followed the SSP, while accounting for possible biases in the availability of certain phonological transcriptions. We also derived a cross-linguistic, global estimate of this prevalence. To examine whether the estimated prevalence of the SSP across languages is also reflected in the speech input that adults and infants receive, in a second analysis we derived a measure of syllabic frequency that we used to estimate how often SSP-following and SSP-violating syllables appear in adult-directed, and infant-directed speech.

1 What proportion of the syllabic repertoire of a language follows the SSP?

In this analysis, we estimated the proportion of the syllabic repertoire of each language that followed SSP. Using Bayesian modelling we derived a quantitative estimate of global and language-specific SSP prevalence, along with their corresponding statistical uncertainty.

1.1 Methods

Raw data, reproducible code, and project documentation are available in the Open Science Framework (https://osf.io/yeqms) and GitHub (https://github.com/NeuroDevCo/ssp-corpus).

1.1.1 Data collection

We used the `wikipron` Python package (version 1.3.3, Lee et al., 2020) to scrape all available phonological transcriptions from all 7,859 languages included in the Glottolog database (Nordhoff & Hammarström, 2011). This resulted in ~1.7 million pronunciations belonging to 900 languages. We filtered out languages with less than 100 phonological transcriptions available. The final dataset containing word-form phonological transcriptions included 731,145 word-forms from 192 languages. Information from each language (e.g., language macrofamily) was extracted using the `lingtypology` Python package (version 1.3.3, Voronov & Forkel, 2019; Moroz, 2017).

1.1.2 Data processing

We segmented word-form phonological transcriptions into syllable candidates where possible. If available, we used explicit syllable boundary markers ("`.`") or implicity syllable markers ("ˈ" for primary stress, "ˌ" for secondary stress). In the absence of syllabic markers, we assumed the word to be monosyllabic. We then classified the segments inside each syllable candidate into consonants (`C`), vowels (`V`), and semi-vowels (`v`). Consecutive sequencies of vowels and semi-vowels (e.g., `V`, `VV`, `Vv`, `VVv`) where classified as vocalic clusters. Syllable candidates with more than one vocalic cluster were flagged as multisyllabic and excluded from the dataset. We also remove duplicated syllables within each language (e.g., if the syllable /ka/ appeared in Spanish words /ˈka.sa/ and /kaˈpo/ in Spanish, it would appear more than once in the Spanish dataset). The final result was a syllabic repertoire for each language. This dataset included 113,760 syllables from 139 languages from 17 distinct language families. To evaluate if each extracted syllable followed or violated the SSP, we assigned a sonority score to each

segment in each syllable in a scale from 1 to 9, according to the panphon Python package (version 0.22.2 Mortensen et al., 2016). Syllables with more than one peak (i.e., local minimum) in the sonority sequence were classified as SSP-violating, and as SSP-following otherwise. Syllables with a flat sonority contour were classifid as SSP-following.

Table 1

Examples of SSP-following and SSP-violating syllables in the dataset. Sonority index contours are shown for each syllable numerically and graphically.

Language	Syllable	Original word	Sonority index	
SSP-following				
Faroese	/aɻ/	altar /ˈaɻ̩.taɹ/	9-6	
Finnish	/berk/	Berg /ˈberk/	2-9-6-1	
Kankanaey	/bal/	balla /balˈla/	2-9-6	
Moksha	/alʲ/	альняка /ˈalʲ.nʲa.kə/	9-6	
Sanskrit	/be:/	कुबेर /ku.be:.ɾɐ/	2-9	
SSP-violating				
Assamese	/skul/	স্কুল /skul/	3-1-8-6	
German	/amts/	Amtsträger /ˈamtsˌtʁe:gɐ/	9-5-1-3	
English	/bi:dz/	beads /bi:dz/	2-8-2-4	
Serbian-Croatian-Bosnian	/bɾ̂:zo/	brzo /bɾ̂:zo/	2-6-4-9	
Mon	/hma/	ဘျာ /hma/	6-5-9	

1.1.3 Data analysis

To model the prevalence of the SSP across languages, we aggregated, for each language in the dataset, the number of syllables that followed the SSP. This resulted in one data point per language. We used a multilevel, Bayesian logistic regression model to estimate the

proportion of syllables that followed or violated the SSP across languages. This model estimated the average probability of the syllables of each language in the dataset following the SSP (θ_j), and global estimate of the same probability (θ).We included a bias correction parameter that adjusted θ_j and θ by the number of observations in each language's dataset: we did not have access to the full syllabic repertoire of each language, as available syllables from languages are the result of convenience sampling (e.g., in Wiktionary word entries are created by volunteers). The number of entries from each language, the amount of detail with which they are entered, and the type of words that are entered, varies systematically across languages. This might bias the estimates. For instance, if SSP-violating syllables are less frequent than SSP-following syllables in the lexicon of a language, and less frequent words are less likely to enter Wiktionary by volunteers, then the observed prevalence of SSP for that language likely overestimates the prevalence of SSP-following languages for a similar discussion (see Yin et al., 2023 for a similar observation). For this reason, this model included a term in the linear model γ/n_i aimed at capturing this potential overrepresentation of SSP-following syllables in datasets with fewer observations (see Appendix B for more details).

Equation 1 shows a detailed implementation of both models model. We implemented the model in the probabilistic programming language Stan (version 2.38, Carpenter et al., 2017) and CmdStanPy, its interface to its Python 3.12.7. We run eight Hamiltonian Monte Carlo chain of 2,000 iterations each (1,000 of which were warm-up iterations) to estimate the posterior distribution of the model.

Likelihood:

Global prevalence of the SSP:

$$y \sim \text{Binomial } (n_j, \frac{\theta}{1-\theta})$$

$$\theta = \mu + (\gamma / n_j)$$

Language-specific prevalence of the SSP:

$$y_j \sim \text{Binomial } (n_j, \frac{\theta_j}{1-\theta_j})$$

$$\theta_j = \mu + u_j + (\gamma / n_j)$$

where:

j indexes each language (one datapoint)

y_j is the number of SSP-following syllables in language j

n_j is the tota number of syllables in language j

u_j is a language-specific adjustment to μ

γ introduces an adjustment to μ that is inverselly proportional to n_i

(1)

Prior:

$$\mu, \gamma \sim \text{Normal } (0, 1)$$

$$u_j \sim \text{Normal } (0, \sigma_u)$$

$$\sigma_u \sim \text{Exponential } (2)$$

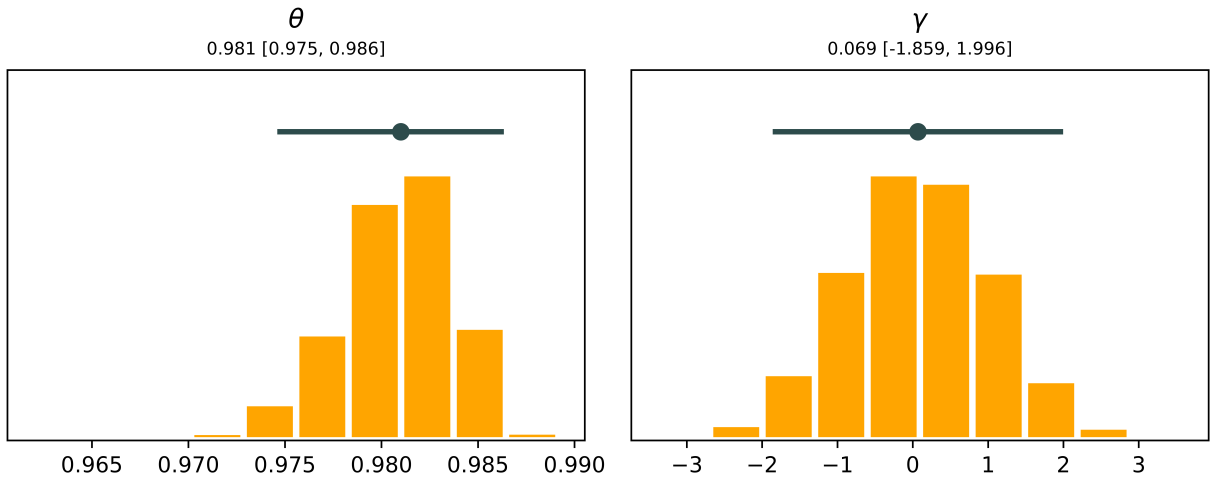
1.2 Results

We report the median and 95% high density interval (HDI) of the posterior distribution of the model parameters of interest. The 95% HDI is the narrowest continuous interval that

contains the 95% of the posterior distribution of the parameter. This interval can be interpreted as being 95% likely to contain the true value of the parameter, given the observed data and the prior distribution of the model (Kruschke & Liddell, 2018).

Figure 2

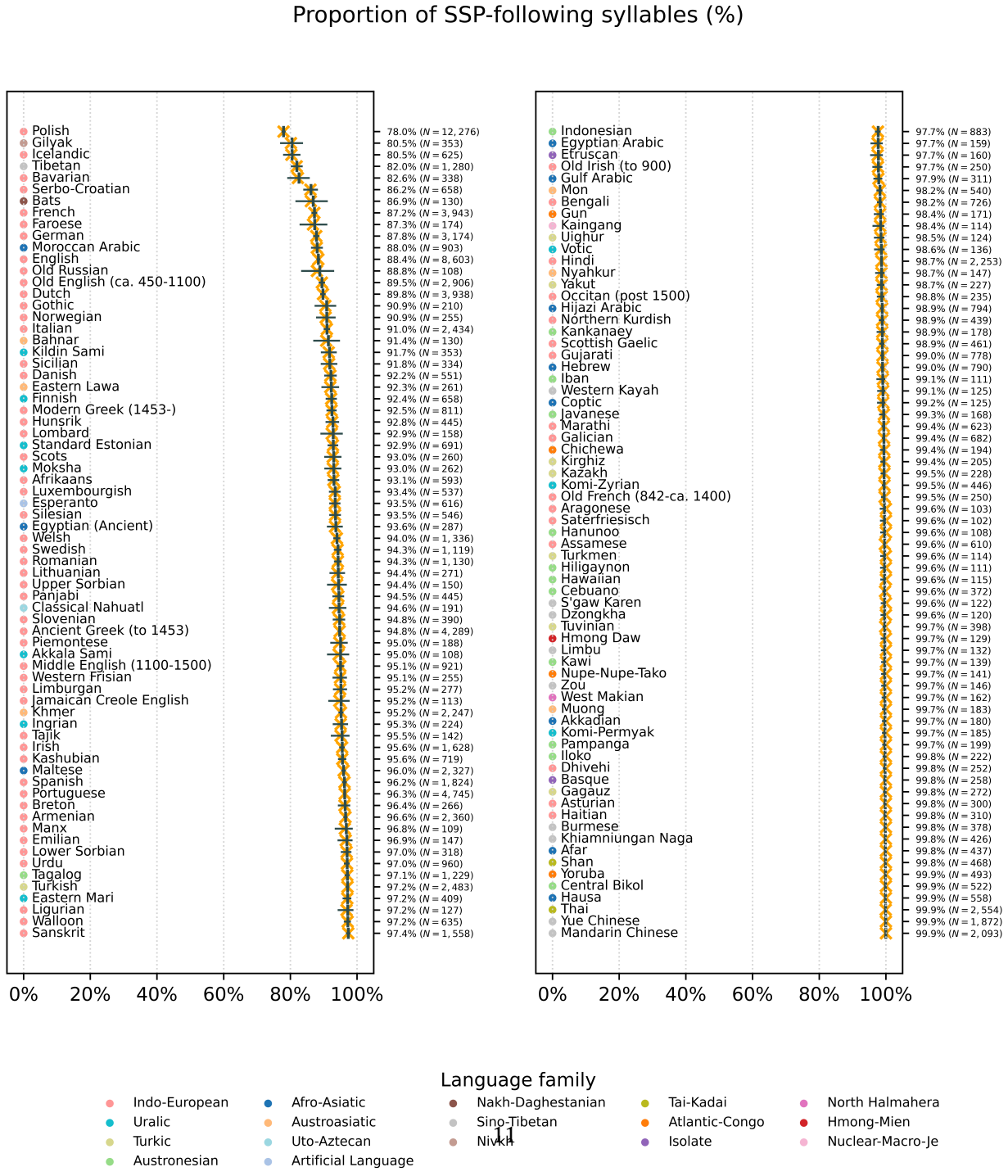
Posterior distribution of parameters of interest θ (cross-language prevalence of SSP-following syllables) and γ (bias parameter) in Equation 1.



The median of the posterior distribution of the cross-language proportion of SSP-following syllables was $\theta = 0.981$ (95% HDI = [0.975, 0.985]), and the language-specific prevalence of the SSP ranged from 77.957% (Polish) to 99.949% (Mandarin Chinese). Only 0.72% of the languages in the sample showed less than 80% of SSP prevalence, 10.79% languages showed less than 90% of SSP prevalence, and 31.65% languages showed less than 95% of SSP prevalence. The median of the posterior distribution of the bias parameter was $\gamma = 0.069$ (95% HDI = [−1.556, 1.666]) (see Figure 1). Figure 3 shows the posterior mean and 95% HDI of the proportion of SSP-following syllables for each language family in the dataset (see Appendix A for a more detailed report).

Figure 3

Posterior, model-estimated proportion of distinct syllables that follow the Sonority Sequencing Principle (SSP) for each language family. Vertical lines and intervals indicate the posterior mean proportion of SSP-following syllables for each language family and its 95% HDI (also numerically reported next to each language). Yellow cross indicate the observed proportion of SSP-following syllables.



2 Are SSP-following syllables more frequent than SSP-violating syllables?

Because items inside the syllabic repertoire of a language likely occur at different rates in speech (e.g., both /ka/ and /gla/ syllables are attested in Spanish, but the former likely appears in a larger number of words), a second analysis investigated whether SSP-following syllables are also more frequent speech compared to SSP-violating syllables. In this second analysis, We explored this research question in two datasets. In the first dataset, syllabic frequencies were derived from available databases of lexical frequencies that did not discriminate between adult-directed speech (ADS) and child-directed speech (CDS). In the second dataset, syllabic frecuencies were derived from lexical frequencies in the CHILDES (MacWhinney & Snow, 1990), which includes corpora of child-directed speech.

2.1 Method

2.1.1 Data collection

We extracted adult lexical frequencies for each word-form in the words dataset using the `wordfreq` Python package (Speer, 2022). This package provides an interface to retrieve lexical frequency data for words in over 40 languages. Lexical frequencies are extracted from a variety of sources according to their availability for each language, including Wikipedia, subtitles (e.g., OPUS Open subtitles 2018 and SUBTLEX), news, books, web text, Twitter, and Reddit, among others. At least three of these sources contribute to the calculation of lexical frequencies in each of the available languages[1]. When lexical frequencies were not available for a particular language (e.g., the ISO 639-3 code does not match any available language in `wordfreq`), an equivalent language was used, if available. Equivalent languages were searched according to their distance to the original language according to CLDR v38.1. This system assigns a numeric value to each pair of language codes based on whether both codes are assigned to identical languages with different names (e.g., Norwegian Bokmål → Norwegian,

[1]Importantly, lexical frequencies provided by `wordfreq` have have not been updated since September 2024 to prevent AI generated text from biasing the data.

score 0), minor regional differences (score 1-3, e.g., Norwegian Bokmål ⟶ Norwegian), and languages with significant but unproblematic regional differences (score 10-24, e.g., Afrikaans ⟶ Dutch, Wu Chinese ⟶ Mandarin Chinese), etc. The rationale behind using equivalent languages to retrieve missing lexical frequencies was that users of close languages are likely to use similar words to refer to the same objects, and therefore lexical frequencies are expected to be reasonably equivalent across such language pairs.

2.1.2 Data processing

For each syllable, we aggregated the lexical frequency (frequency per million words) of each word the syllable was part of in the dataset. The rationale behind this step is that each time a word is uttered, each of its syllables is also uttered. Therefore, by calculating the lexical frequency of each word a given syllable is involved in, an estimation of syllabic frequency can be derived. Table 2 provides illustrative examples of how syllabic frequencies were calculated. The final dataset for ADS syllabic frequencies comprised 52,588 syllables from 42 languages from 8 distinct language families. For syllabic frequencies in CDS, the number of syllables in the dataset was 21,276 from 17 languages from 3 distinct language families.

Table 2

Illustrative example of syllabic frequency calculation for syllables /ka/ and /ne/ across French and Spanish. PMW*: Frequency per million words.* Zipf*: Zipf-transformed frequency score.*

	French			Spanish		
	Word	Adult	CHILDES	Word	Adult	CHILDES
Syllable /ka/						
	/ka.li.te/	131.83	0.30	/kaˈfe/	64.57	124.73
	/vɔ.ka.sjɔ̃/	16.98	0.30	/kaˈsiʝa/	5.62	4.88
	/vɔd.ka/	4.37	1.52	/kaˈxita/	2.45	16.87
	/ka.di.jak/	2.00	0.30	/kaˈlada/	0.65	0.89
	/ka.na.pe/	0.87	1.52	/kaˈbaʝa/	0.37	0.89
PMW		156.04	3.96		73.66	148.25
Zipf		5.19	3.60		4.87	5.17
Syllable /ne/						
	/ne/	61.66	8.53	/teɾmiˈne/	9.77	5.33
	/ʁɛ̃.dɔ.ne/	6.92	0.30	/neˈbeɾa/	4.27	11.10
	/de.kli.ne/	2.24	0.30	/neˈbado/	1.45	11.10
	/ku.ʁɔ.ne/	1.91	0.30	/neˈnita/	0.47	1.78
	/ʁe.zɔ.ne/	1.17	0.61	/neˈgadas/	0.25	0.44
PMW		73.90	10.06		16.20	29.74
Zipf		4.87	4.00		4.21	4.47

2.1.3 Data analysis

We used a multilevel, Bayesian, distributional regression model to estimate the proportion of syllables that followed or violated the SSP across languages. The response variable (Frequency) was modelled using a log-normal distribution, with a location parameter μ and a scale parameter σ. We modelled both parameters using linear regression models, each with a global intercept (β_0, π_0) with a language-specific adjustment (u_{0j}, w_{0j}), a predictor variable SSP, which indexed whether the syllable followed the SSP (coded as `+0.5`) or violated the SSP (coded as `-0.5`) and its corresponding regression coefficient (β_1, π_1), with language-specific adjustments (u_{1j}, w_{1j}). Equation 2 provides a formal definition of the model. We

specified weakly informative priors for the parrameters in the model. We run eight Hamiltonian Monte Carlo chain of 2,000 iterations each (1,000 of which were warm-up iterations) to estimate the posterior distribution of the model.

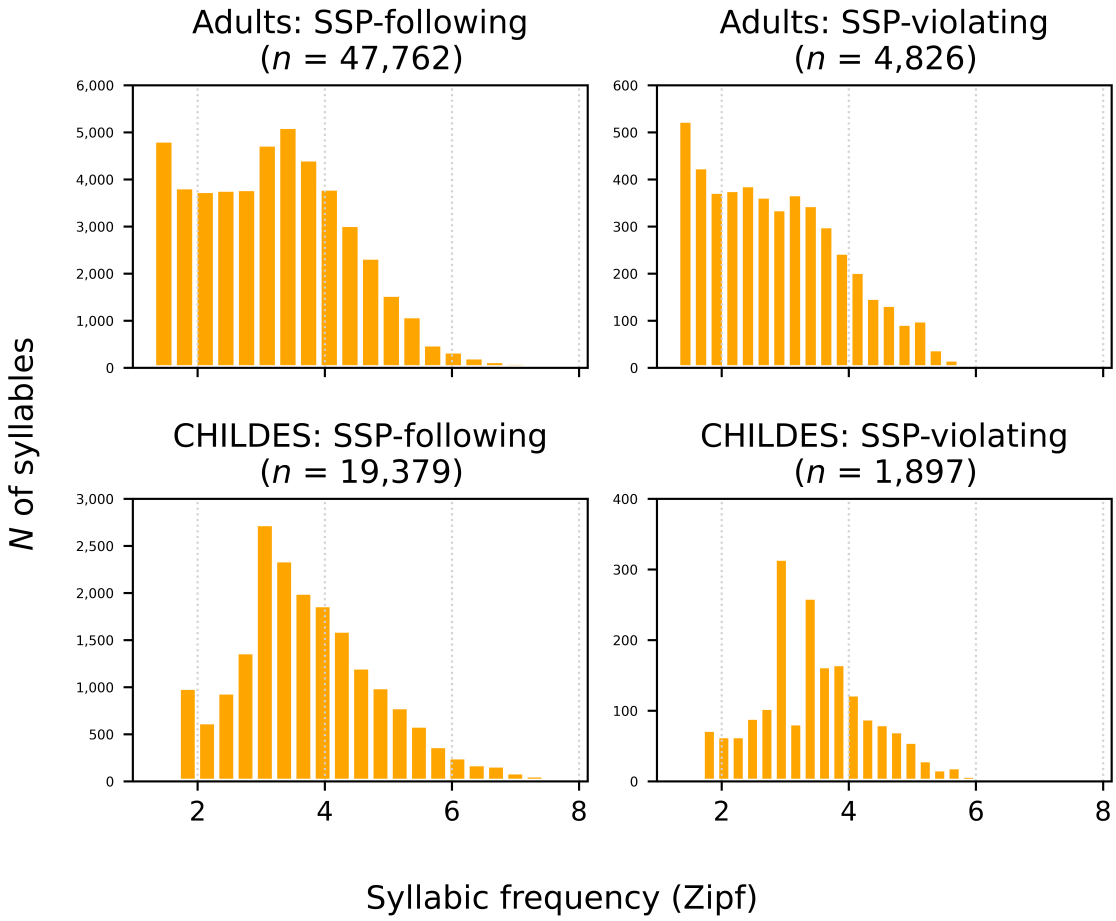
$$
\begin{aligned}
&\textbf{Likelihood}: \\
&y_{ij} \sim \text{LogNormal } (\mu_{ij}, \sigma_{ij}) \\
&\mu_{ij} = \beta_0 + u_{0j} + (\beta_1 + u_{1j}) \cdot \text{SSP}_{ij} \\
&\sigma_{ij} = \exp(\pi_0 + w_{0j} + (\pi_1 + w_{1j}) \cdot \text{SSP}_{ij}) \\
&\\
&\textbf{Prior}: \\
&\beta_{0,1} \sim \text{Normal } (0, 10) \\
&\pi_{0,1} \sim \text{Normal } (0, 5) \\
&u_{0,1j} \sim \text{Normal } (0, \tau_{uj}) \\
&w_{0,1j} \sim \text{Normal } (0, \tau_{wj}) \\
&\tau_{uj}, \tau_{wj} \sim \text{Exponential } (2)
\end{aligned}
\tag{2}
$$

2.2 Results

Figure 4 shows the distribution of syllabic frequencies across languages for SSP-following and SSP-violating syllables. The posterior distribution of the model parameters is shown in Figure 5. The results from the Bayesian model in adult frequencies indicate that the 95% HDI of the difference between SSP-following and SSP-violating syllables in the location parameter exceeded zero (μ = 0.024, 95% HDI = [0.003, 0.044]), pointing to SSP-following syllables being more frequent than SSP-violating syllables across languages.

Figure 4

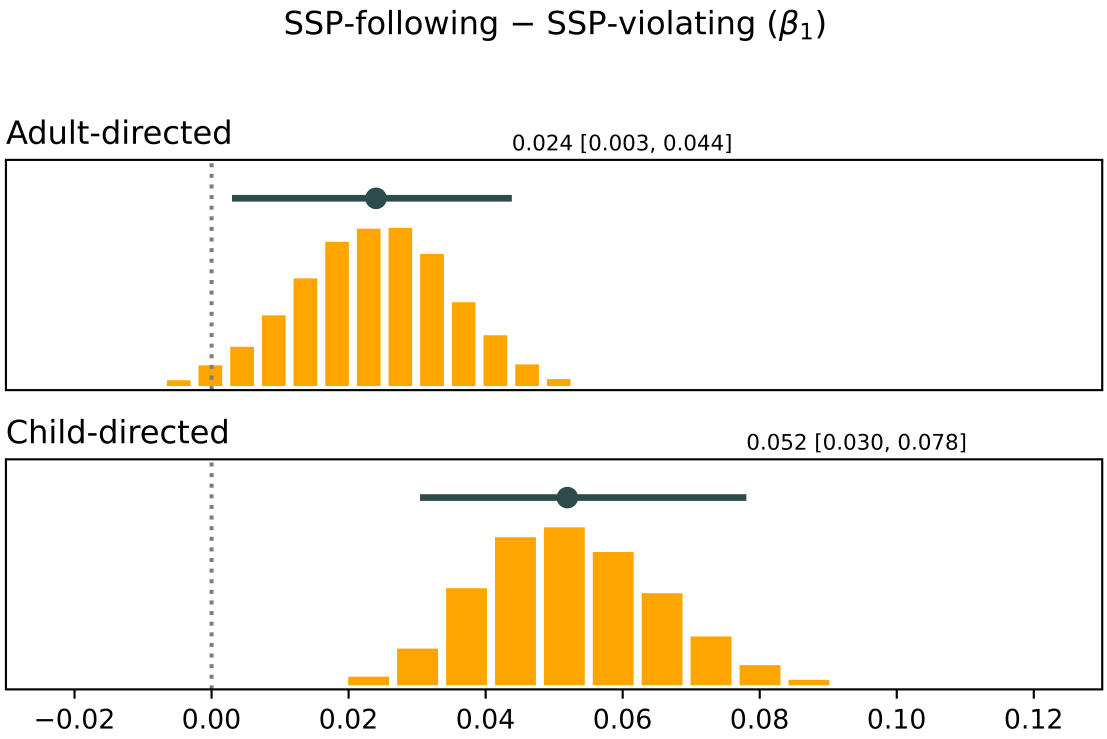
Distribution of syllabic frequencies across languages for SSP-followingg and SSP-violating syllables.



We observed a similar pattern in CDS syllabic frequencies. In this dataset, the posterior distribution of the difference between SSP-following and SSP-violating syllables in the location parameter also exceed zero (μ = 0.052, 95% HDI [0.030, 0.078]), indicating that SSP-following syllables are more frequent than SSP-violating syllables across languages.

Figure 5

Syllabic frequencies in adult-directed and child-directed speech.



3 Discussion

The present study was aimed at quantifying the cross-linguistic prevalence of the sonority sequencing principle (SSP). We gathered gathered a large number of phonological transcriptions from the Wiktionary database, and derived a syllabic repertoire for each of the 139 languages from 17 language families. We evlauated the proportion of SSP-following syllables present in each syllabic repertoire, and used Bayesian modelling to estimate the cross-lingusitic, and language-specific prevalence of the SSP. We then derived a measure of syllabic frequency for each syllable in the dataset, and compared the relative frequency with which SSP-following and SSP-violating syllables appeared in adult-directed speech (ADS) and child-directerd speech (CDS). Our results show that an overwhelming majority oof the syllables across languages follow the SSP, with very few languages showing less than 90% prevalence of

the SSP in their syllabic repertoire. We also found that SSP-following syllables were more frequent than SSP-violating syllables in both ADS and CDS.

0) Recap of study motivation and aims. Highlight syllabic frequency measure (novely compared to previous literature).
1) Summary and interpretation of results: 1.1) Very high observed prevalence of SSP across languages. 1.2) Very few languages show less than 80% of SSP prevalence 1.3) Evidence that SSP-following syllables are slightly more frequent than SSP-violating syllables, both in CDS and ADS.
2) Comparison with previous estimates of SSP, with especulation about why results may differ: 2.1) Yin et al. (2023): they show that virtually all languages show violations of the SSP. We provide converging evidence, while emphasizing that, while such violations exist, they may play a marginal role in language acquisition. 2.2) Other studies showing lower prevalence of the SSP within or across languages: 2.2.1) Different data sources: phonological inventories of attested consonant clusters vs. syllabic repertoires, syllabic frequencies. 2.2.2) Possible bias in our dataset: discuss how we accounted for the dataset size bias, and how if differs from previous attempts to correct for it.
3) Implications: 2.1) Evidence that the SSP may be available as a cue for syllabification (specially) during language acquisition 2.2) The present status does not argue in favour or against the universality status of the SSP, but does point to the SSP as a universally available, functionally relevant phonological feature
4) Limitations: 4.1) No direct evidence of SSP prevalence in the context of language usage (ideally, we would transcribe and syllabify a corpus and simply calculate the proportion of SSP-following syllables). Currently not possible (few datasets are syllabified). 4.2) Althoug the present study includes a relatively large sample of languages from different language

families, Indoeuropean (mostly Western) languages are overrepresented in the dataset, paralelling the same pattern observed in corpus analysis in Linguistics.

5) Conclusions

4 Appendix A

5 Number of SSP-following and SSP-violating syllables per language and language family

Table A1

Number of SSP-following and SSP-violating syllables per language and language family.

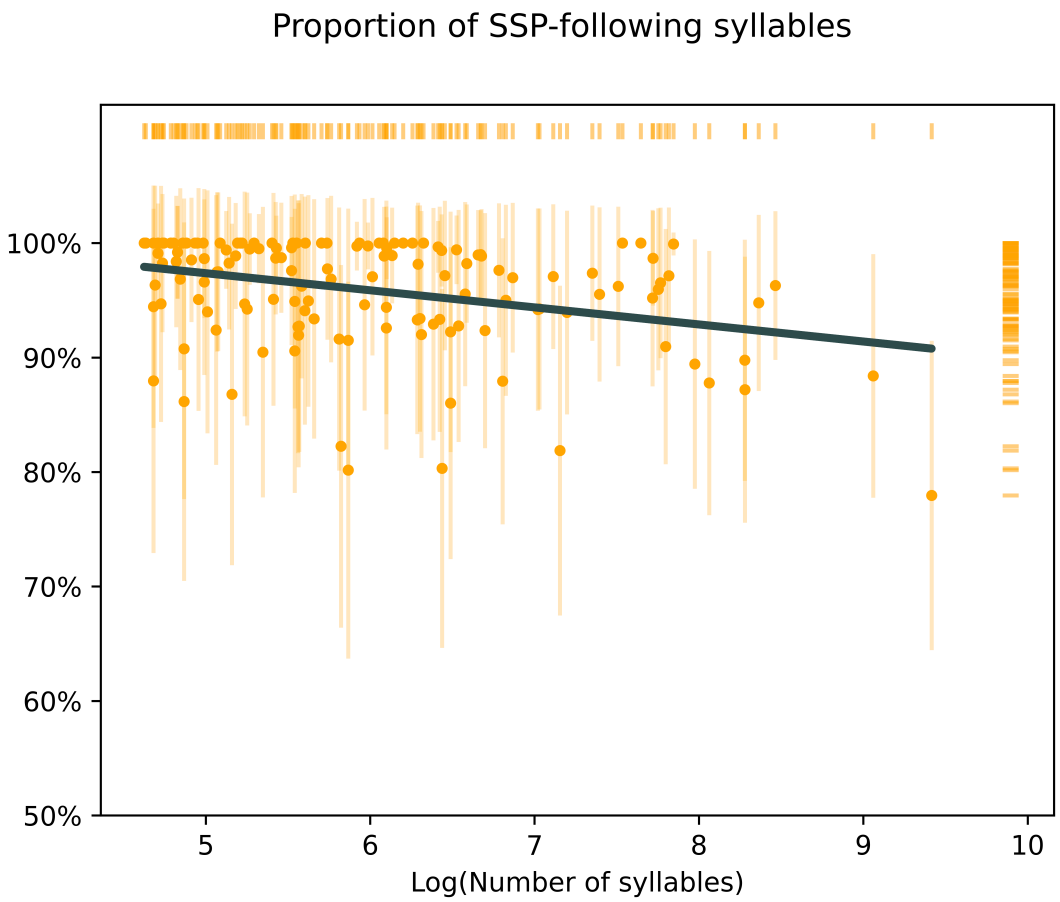
	N syllables	N words	Following	Violating	% Following
Afro-Asiatic					
Moroccan Arabic	903	833	794	109	87.93%
Egyptian (Ancient)	287	942	268	19	93.38%
Maltese	2,327	3,777	2,233	94	95.96%
Egyptian Arabic	159	148	155	4	97.48%
Gulf Arabic	311	343	304	7	97.75%
Hijazi Arabic	794	952	785	9	98.87%
Modern Hebrew	790	948	782	8	98.99%
Coptic	125	128	124	1	99.20%
Hausa	558	738	558	0	100.00%
Afar	437	1,335	437	0	100.00%
Akkadian	180	184	180	0	100.00%
Artificial Language					
Esperanto	616	5,975	575	41	93.34%
Atlantic-Congo					
Gun	171	143	168	3	98.25%
Nyanja	194	197	193	1	99.48%
Yoruba	493	1,311	493	0	100.00%
Nupe-Nupe-Tako	141	153	141	0	100.00%
Austroasiatic					
Bahnar	130	198	118	12	90.77%
Eastern Lawa	261	254	240	21	91.95%
Central Khmer	2,247	6,070	2,139	108	95.19%
Mon	540	1,051	530	10	98.15%
Nyahkur	147	152	145	2	98.64%
Muong	183	195	183	0	100.00%
Austronesian					
Tagalog	1,229	13,663	1,193	36	97.07%
Standard Indonesian	883	3,851	862	21	97.62%
Kankanaey	178	422	176	2	98.88%
Iban	111	254	110	1	99.10%
Javanese	168	145	167	1	99.40%
Cebuano	372	1,556	371	1	99.73%
Kawi	139	218	139	0	100.00%
Asturian-Leonese-[illegible] Cantabrian	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

6 Appendix B

7 Relationship between number of observations and SSP prevalence

Figure B1

Relationship between the number of observed syllables and the SSP-prevalence in each language. The X-axis indicates the log-transformed number of observed syllables in the dataset. The Y-axis indicates the SSP prevalence (i.e., observed proportion of syllables that follow the SSP in the language's dataset). Each dot and interval indicate the data for one language, and the standard error of the SSP-prevalence. Marks in the top and right margin of the figure indicate the distribution of the observations. The predictions of a linear regression model between both variables are shown as a green line.



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